

ĐỀ BÀI: ARTWORK MANAGEMENT SYSTEM

Yêu cầu chung:

- URL: http://localhost:8084/ArtGallery_System/
- Công nghệ: Java Servlet, JSP, JDBC, SQL Server.

1. Database Script (SQL Server)

```
CREATE DATABASE PaintingDB;
```

```
GO
```

```
USE PaintingDB;
```

```
GO
```

```
CREATE TABLE tblUsers (  
    userID varchar(20) PRIMARY KEY,  
    fullName nvarchar(50) NOT NULL,  
    password varchar(50) NOT NULL,  
    roleID varchar(5), -- 'AD' hoặc 'US'  
    status int -- 1: Active, 0: Inactive  
);
```

```
CREATE TABLE tblPaintings (  
    paintingID varchar(10) PRIMARY KEY,  
    description nvarchar(255),  
    creator nvarchar(50),  
    height float,  
    width float,  
    price money,  
    isAvailable bit -- 1: Còn hàng, 0: Đã đặt trước/Hết hàng  
);
```

INSERT INTO tblUsers VALUES

('admin1', 'The Boss', '1', 'AD', 1),

('user1', 'Customer A', '1', 'US', 1),

('baduser', 'Locked User', '1', 'US', 0);

INSERT INTO tblPaintings VALUES

('P1', 'Ocean Blue', 'Monet', 50, 70, 1000, 1),

('P2', 'Dark Night', 'Gogh', 60, 60, 0, 1),

('P3', 'Modern Art', 'Unknown', 40, 40, 500, 0);

2. Các yêu cầu chi tiết

No	Function	Detailed Logic & Constraints	Mark
1	Login Logic	Distinguish between 3 specific error cases: - Wrong ID: "User does not exist". - Wrong Pass: "Incorrect Password". - Inactive: "This account is not activated". -if login succuss, go to home.jsp	1.5
2	Role & Welcome	- Welcome: Display Welcome fullName (roleID: ADMIN CUSTOMER) on the all page after login(home, search, update page). - Have 2 link: search and logout. This link will also display on search page. - UI Control: The "Update" button/link must only appear for roleID = 'AD'.	1.0
3	Logout	Destroy session and redirect to login.jsp.	1.5

4	Search	- If search box is empty, show no results (do not show the table). - Filter: Do not display items where Price <= 0. - Column: Show Status as "Available" (isAvailable=1) or "Sold Out" (isAvailable=0).	3.0
5	Update & Security	- Role Security: Only the AD role sees the Update button. - Update field: - description nvarchar(255), - creator nvarchar(50), - height float, - width float, - price money - Bypass Protection: If a US user types the URL manually, redirect to paintingList.jsp and show: "No Permission" in a red tag. - Component: isAvailable must be a Checkbox. Handle the error and show the appropriate error to the user. - When update successfully, redirect to search page with search grid.	3.0